

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EE) Examination

May 2014

EE-414 Power System Operations

Date: 24.05.2014, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 What is the purpose of contingency analysis? Explain Contingency analysis in detail. [07]
- Q - 2 (a) Draw typical bath tub curve. Explain the three regions of the curve. For which region are reliability calculations valid? Why? [07]
- (b) Explain load forecasting methodology and estimation of average and trend terms. [07]

OR

- Q - 2 (a) Explain the term 'Reliability', 'Outage', 'MTTF', 'MTBF'. Also give difference between forced and scheduled outage. [07]
- (b) Explain Kalman Filtering approach for short term forecasting. [07]
- Q - 3 (a) Draw the DSM planning stages diagram. Also explain each of the stages in detail. [07]
- (b) What is Continuous Markov process? Use continuous markov process to find the probabilities of being available and not being available for a single repairable component having failure rate λ and repair rate μ . [07]

OR

- Q - 3 (a) What is the scope of demand side management? Explain the types of DSM measures and programs. [07]
- (b) Define the term Failure rate and Repair rate. Also derive the expression of probabilities of survival and failure. [07]

SECTION – II

- Q - 4 What are the different entities in deregulated environment? [07]
- Q - 5 (a) What are the benefits and challenges of distribution automation system? [07]
- (b) What do you mean by deregulation? Explain structure of deregulated industry. [07]

OR

- Q - 5 (a) Define the term Distribution Automation. Explain the primary functions of DA. [07]
- (b) What are reasons for restructuring? Explain each of them in details. [07]

Q - 6 (a) Explain application of power system state estimation. [07]

(b) Explain Lambda-Iteration method of Economic dispatch problem in detail. [07]

OR

Q - 6 (a) Describe Least Square Estimate (LSE) and weighted LSE for estimation of x vector of n variable using another vector y of variable m where $m > n$. [07]

(b) Two units of the system have the following cost curves, [07]

$$F_1 = 120 + 22P_{G1} + 0.05P_{G1}^2 \text{ Rs/h}$$

$$F_2 = 120 + 16P_{G2} + 0.06P_{G2}^2 \text{ Rs/h}$$

Both the units operate at all times and the maximum and minimum loads on each units are 100 MW and 20 MW, respectively. Determine the economic operating schedule of the plants for loads of 80 MW, 120 MW, and 180 MW. Neglecting the transmission line losses.
